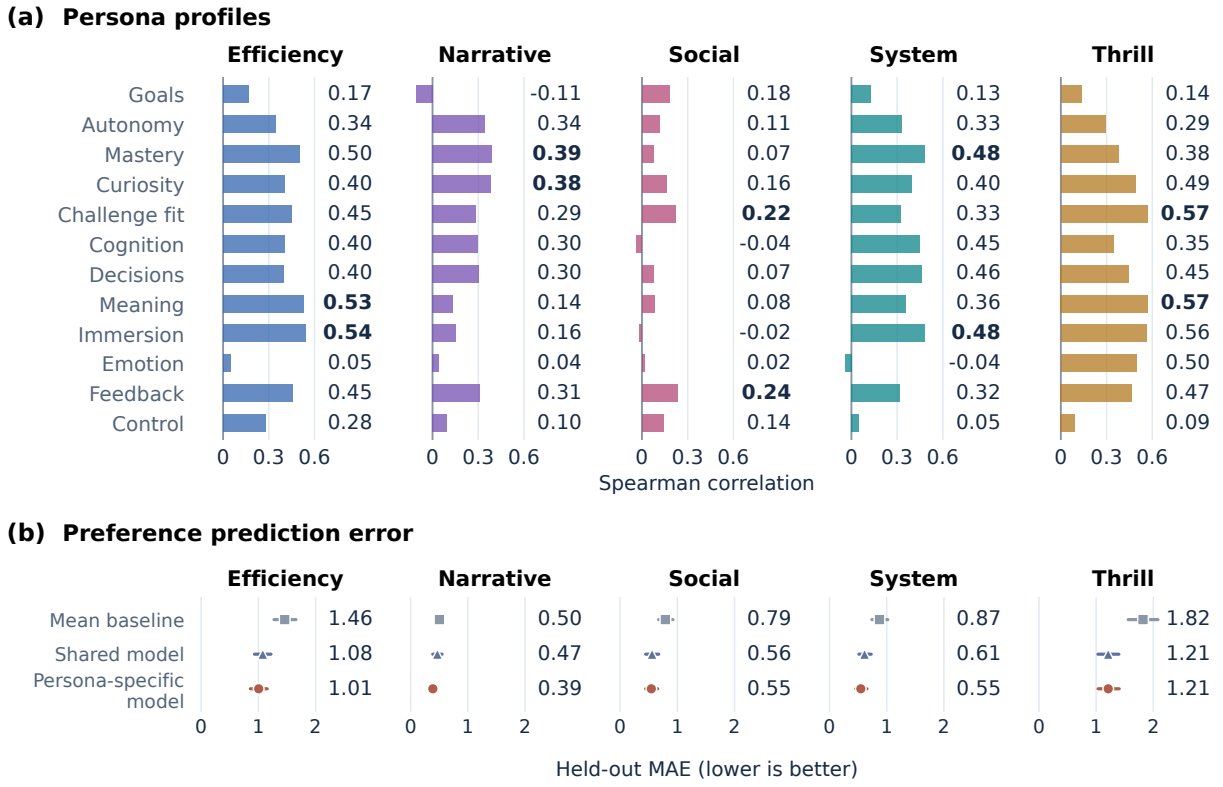
(a) Persona profiles
Efficiency
Narrative
Social
System
Thrill
Goals 0.17 -0.11 0.18 0.13 0.14
Autonomy 0.34 0.34 0.11 0.33 0.29
Mastery 0.50 0.39 0.07 0.48 0.38
Curiosity 0.40 0.38 0.16 0.40 0.49
Challenge fit 0.45 0.29 0.22 0.33 0.57
Cognition 0.40 0.30 -0.04 0.45 0.35
Decisions 0.40 0.30 0.07 0.46 0.45
Meaning 0.53 0.14 0.08 0.36 0.57
Immersion 0.54 0.16 -0.02 0.48 0.56
Emotion 0.05 0.04 0.02 -0.04 0.50
Feedback 0.45 0.31 0.24 0.32 0.47
Control 0.28 0.10 0.14 0.05 0.09
0 0.3 0.6
Spearman correlation
(b) Preference prediction error
Efficiency
Narrative
Social
System
Thrill
Mean baseline 1.46 0.50 0.79 0.87 1.82
Shared model 1.08 0.47 0.56 0.61 1.21
Persona-specific model 1.01 0.39 0.55 0.55 1.21
0 1 2
Held-out MAE (lower is better)